

HORSHAM, AustraliaWMO: **958390**Lat: **36.6697S**Lon: **142.1731E**Elev: **134**StdP: **99.72**Time Zone: **10.00 (AUS)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.5	0.7	-5.0	2.5	24.8	-2.9	3.0	19.6	13.0	12.0	12.0	12.1	2.7	80	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.3	38.0	18.3	35.5	17.7	33.2	17.3	20.8	29.3	19.7	29.0	18.8	28.7	6.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	22.2	17.1	12.4	21.7	15.6	11.3	21.1	60.6	29.6	56.9	28.4	53.8	28.6	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	11.0	9.9	DB	-2.7	42.8	0.9	1.8	-3.3	44.1	-3.9	45.1	-4.3	46.1	-5.0	47.4	(4)
(5)			WB	-3.0	22.5	0.9	1.3	-3.6	23.5	-4.1	24.2	-4.6	24.9	-5.2	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	21.5	21.0	18.5	14.9	11.4	8.9	8.4	8.9	10.7	13.4	16.9	19.4	(6)	
	DBStd	5.78	4.36	4.04	3.96	3.24	2.54	2.00	1.94	2.19	2.68	3.71	4.30	4.35	(7)	
	HDD10.0	191	0	0	0	2	13	43	56	49	21	6	1	0	(8)	
	HDD18.3	1797	14	16	47	113	216	282	308	293	231	160	79	38	(9)	
	CDD10.0	1819	356	309	264	148	56	11	7	14	41	113	209	291	(10)	
	CDD18.3	381	112	91	53	9	0	0	0	0	1	8	37	70	(11)	
	CDH23.3	6296	1788	1323	789	184	7	0	0	1	22	247	689	1244	(12)	
	CDH26.7	3154	993	675	356	52	0	0	0	0	2	93	322	661	(13)	
(14) Wind	WSAvg	5.1	5.6	5.4	5.1	4.6	4.7	4.6	5.0	5.0	5.0	5.4	5.5	5.7	(14)	
(15) Precipitation	PrecAvg	443	26	21	21	28	46	45	52	51	50	46	35	23	(15)	
	PrecMax	761	88	111	60	105	125	145	113	98	107	183	92	79	(16)	
	PrecMin	182	1	0	0	1	8	6	14	7	11	5	5	0	(17)	
	PrecStd	114	22	20	15	26	28	28	25	22	27	34	22	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	39.5	37.2	31.5	24.1	17.7	17.0	21.2	26.2	33.1	36.6	40.6	(19)	
		MCWB	19.4	20.2	17.8	16.3	13.6	11.8	11.4	12.6	13.9	15.6	17.5	17.7	(20)	
	2%	DB	38.8	36.1	33.4	27.8	20.9	15.8	15.1	17.8	22.3	29.4	33.4	36.8	(21)	
		MCWB	18.5	18.5	17.7	15.3	13.1	11.3	10.9	11.6	13.5	14.6	16.5	17.5	(22)	
	5%	DB	35.3	33.7	30.7	25.3	18.6	14.6	13.8	15.6	19.4	25.9	30.8	33.5	(23)	
		MCWB	17.9	18.0	16.8	14.6	12.7	11.0	10.4	10.9	12.3	14.1	16.1	16.5	(24)	
	10%	DB	32.2	31.0	27.8	22.7	16.8	13.5	12.7	14.0	17.0	22.6	27.4	30.3	(25)	
		MCWB	17.2	17.2	16.3	13.8	12.3	10.6	9.9	10.2	11.6	13.3	15.2	16.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	22.5	20.2	17.7	15.7	13.5	12.7	13.7	15.9	18.0	20.1	21.2	(27)	
		MCDB	30.3	33.1	28.3	24.8	18.9	15.6	14.9	18.2	21.6	24.9	29.1	27.6	(28)	
	2%	WB	20.7	20.9	19.1	16.6	14.5	12.4	11.7	12.3	14.3	16.3	18.7	19.2	(29)	
		MCDB	29.8	29.3	27.6	23.5	18.4	14.6	13.9	16.4	19.9	23.9	27.9	28.1	(30)	
	5%	WB	19.6	19.6	18.0	15.6	13.7	11.6	11.0	11.4	13.1	15.0	17.5	18.1	(31)	
		MCDB	30.5	28.5	27.3	22.4	17.3	13.8	13.0	14.7	18.3	23.6	25.7	29.3	(32)	
	10%	WB	18.4	18.4	17.0	14.6	12.8	10.9	10.3	10.6	12.0	13.9	16.2	17.0	(33)	
		MCDB	30.1	28.2	26.5	21.2	16.1	12.9	12.3	13.5	16.5	21.7	25.1	28.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	16.6	15.5	14.3	11.4	9.8	9.3	11.0	13.0	15.5	16.4	16.9	(35)	
		MCDBR	21.8	20.2	19.5	18.3	14.3	11.4	10.5	13.6	17.3	21.7	21.6	22.1	(36)	
	5% WB	MCWBR	7.3	6.8	7.4	8.3	8.2	7.5	7.0	8.4	9.4	9.5	7.7	7.4	(37)	
		MCWBR	16.5	15.8	16.3	14.6	11.1	9.7	9.0	11.9	14.5	17.8	17.3	16.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.338	0.333	0.316	0.309	0.300	0.292	0.292	0.304	0.322	0.331	0.337	0.330	(40)	
		taud	2.497	2.520	2.566	2.566	2.577	2.593	2.592	2.535	2.473	2.457	2.473	2.512	(41)	
	Ebn at Noon		996	978	958	905	855	835	856	896	934	969	991	1009	(42)	
		Edh at Noon	114	106	94	82	71	65	69	83	101	112	116	113	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.69	6.83	5.42	3.84	2.58	2.06	2.22	3.13	4.33	5.77	6.81	7.55	(44)	
	RadStd		0.40	0.33	0.25	0.20	0.12	0.15	0.13	0.20	0.31	0.41	0.43	0.36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+97	(46)
(47)	Regional (27 neighbors)	+0.43	N/A	N/A	+0.86	N/A	N/A	N/A	-87	+139	+59	(47)

Nomenclature: See separate page